



Hybrid Method to Explain Predictions of Stacking Ensemble Model

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Received: 18 October 2024 / Accepted: 4 December 2025 / Published online: 7 January 2026
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Abstract

The contribution of artificial intelligence (AI) to solving real-world problems is well established, but the issue of trusting AI systems remains an ongoing challenge. AI systems are considered to be complex because of their decision-making mechanism, which are not easily understandable by users. The stacking model is one of these complex models that is widely used because it offers better predictive performance. However, many studies focus solely on exploiting predictive performance without paying much attention to understanding the predictions of the stacking model. Nevertheless, among the few studies that have addressed this aspect of explainability, some have used existing agnostic methods to provide explanations, while others provide explanations based on a subset of base learners rather than the entire stacking model. In this work, a hybrid method is proposed and designed to explain the predictions of stacking models based on their internal behavior. The proposed method combines the Layer-wise Relevance Propagation (LRP) with model-agnostic feature importance techniques, such as SHapley Additive exPlanations (SHAP) or Local Interpretable Model-agnostic Explanations (LIME). Experimental results show that the method consistently exhibits greater stability than both SHAP and LIME, and in certain cases, provides higher fidelity relative to the size of the explanation.

Keywords Explainable artificial intelligence · Stacking model · Stacking ensemble model · Stacked generalization · Machine learning

1 Introduction

Artificial intelligence is currently one of the most widely used technologies in our society, thanks to the impressive performance demonstrated by machine learning algorithms regarding various problems over the years (Guidotti, 2021). Machine learning algorithms have produced some very interesting results in terms of decision support in a number of areas, which has led to their widespread use in almost every field today, including health, environment, ecology,

economy, justice, banking, digital technology, and more (Adadi and Berrada, 2018; Ali et al., 2023; Guidotti, 2021; Hajek et al., 2023; Piccialli et al., 2021).

Various categories of machine learning algorithms can be distinguished, including ensemble learning models. Ensemble learning models are based on the idea that combining several models generally produces better results than using a single model (Mienye and Sun, 2022; Azri et al., 2023). In this category, several machine learning models are used to predict individually, and then the resulting predictions are combined to provide the final output. Ensemble models are proposed to further reduce generalization errors during prediction. They often produce more accurate, robust, and better performance but require a lot of computing time. A particular type of ensemble learning model is the stacked generalization ensemble model, also called the stacking model (Sewell, 2011; Ting and Witten, 1999). In this type of model, base models can be of different types, and their results are pooled by another machine learning model called a meta-learner. The meta-learner learns how to better combine the predictions of base models to obtain the final output. The stacking model offers the advantage of being able

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to combine several models of different types that are already performing well individually to obtain even better performance. This is why it is used in various domains, including healthcare (Almohimeed et al., 2023; Baradaran Rezaei et al., 2023; Elakkiya et al., 2021; Lu and Qiu, 2023; Saadatmand et al., 2023; Zhu et al., 2022), industry (Hasan et al., 2024; Kang and Kang, 2021; Liao et al., 2022; Pan and Stark, 2022; Zhang et al., 2023), social networking (Chen et al., 2017, 2022; Liu et al., 2018), education (Hoq et al., 2023), cybersecurity (Gungor et al., 2022), environment (Nguyen et al., 2021), and finance (Papouskova and Hajek, 2019). However, this gain in performance also leads to even greater model complexity and this complexity makes the decision-making process for this type of model difficult for humans to understand.

However, current trends and new regulations in the European Union (General Data Protection Regulation: GDPR) (Mendelson, 2018) suggest that, in addition to good performance, the decision-making logic of algorithm-based systems should be explained to users so that they can understand and trust these systems. In general, it is important to explain how a machine learning model produces a given output decision (Tchuente et al., 2024; Ferretini et al., 2022), because these decisions can have direct consequences on the lives of individuals and users, and it would also make it easier to maintain and improve these intelligent systems (Ali et al., 2023). It is therefore necessary to make the decision-making logic of machine learning models, and even more so ensemble learning models, understandable to humans, by providing appropriate explanations. In the specific case of stacking models, this work focuses on how to effectively explain the predictions of this type of model.

Despite this need to understand the decisions of a machine learning model, several studies (Baradaran Rezaei et al., 2023; Chen et al., 2022, 2017; Elakkiya et al., 2021; Gungor et al., 2022; Hasan et al., 2024; Kang and Kang, 2021; Liao et al., 2022; Liu et al., 2018; Papouskova and Hajek, 2019; Saadatmand et al., 2023; Zhang et al., 2023) that use a stacking model do not address the aspect of explainability even though this model produces good performance.

However, some studies have been carried out to explain the predictions of a stacking model. In some approaches, explanations are derived directly from part of the base models within a stacking ensemble, such as decision trees or linear regressions. While these models are intrinsically interpretable, they don't represent the final prediction, which is made by the meta-learner—raising concerns about the reliability of the explanation if the base models perform poorly. In contrast, other approaches apply model-agnostic explanation techniques such as Local Interpretable Model-agnostic Explanations (LIME) (Ribeiro et al., 2016) or SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017)

to the whole stacking model. These explainability methods consider the model as a black box, and explanations are therefore provided without taking into account the internal workings of ensemble learning models. However, ensemble models are essentially made up of different models, and each of them contributes distinctly to the final result. Not taking their specific architecture into account could result in a loss of information in the process of explaining the predictions. It would be interesting to exploit their architecture to explain the predictions.

Taking into account the internal workings of a stacked generalization model could improve the quality of explanations for predictions. As a more efficient approach, we want to take advantage of both intrinsic and post-hoc explanations. For that, we propose a hybrid explainability method that combines two methods widely used in the literature: the intrinsic (model-specific) Layer-wise Relevance Propagation (LRP) method (Bach et al., 2015), and a model-agnostic method such as SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017). LRP is a method that uses the internal structure of a neural network model to provide explanations for predictions by redistributing the prediction between neurons, starting from the output layer of the network and back-propagating to the input layer. SHAP, on the other hand, is a method that does not consider the internal workings of the model to be explained. It is based on the Shapley value, which is calculated by considering all possible combinations of variables and measuring how the prediction changes when a variable is included or excluded. This method can be used on any type of model.

A similarity of operation can be established between neural networks and stacking models which guided us to adapt LRP to correspond to the workings of the stacking model and the SHAP method to manage the diversity of stacking model components (base models) that can be found in its architecture. The proposed method takes into account the specific operation of a stacking ensemble model to explain the predictions provided by the latter. The idea is to use the concept of level-wise-propagation to deduce the contributions of the inputs from the output of the model. We adapt LRP method and incorporate the SHAP method in the process to explain the predictions of a stacking model.

The highlights of this paper are as follows:

- The proposal of a hybrid method to explain the predictions of a stacking model while taking into account its internal workings.
- An improvement in the stability and fidelity of the stacking model explanations as a result of the previous contribution.
- Numerous experiments to explain the predictions of a stacking model with SHAP, LIME and the proposed

method, followed by a comparison of their results using objective metrics.

The remainder of the paper is organized as follows: Section 2 presents existing studies on explaining the predictions of a stacked generalization ensemble learning model and existing explainability methods. Section 3 presents the proposed method for explaining the predictions of a stacked generalization ensemble learning model. Section 4 presents a summary of the experiment carried out and of the results obtained. Section 5 provides a discussion addressing the implications of this work and outlining potential avenues for future research. Section 6 concludes the study.

2 Background and Literature Review

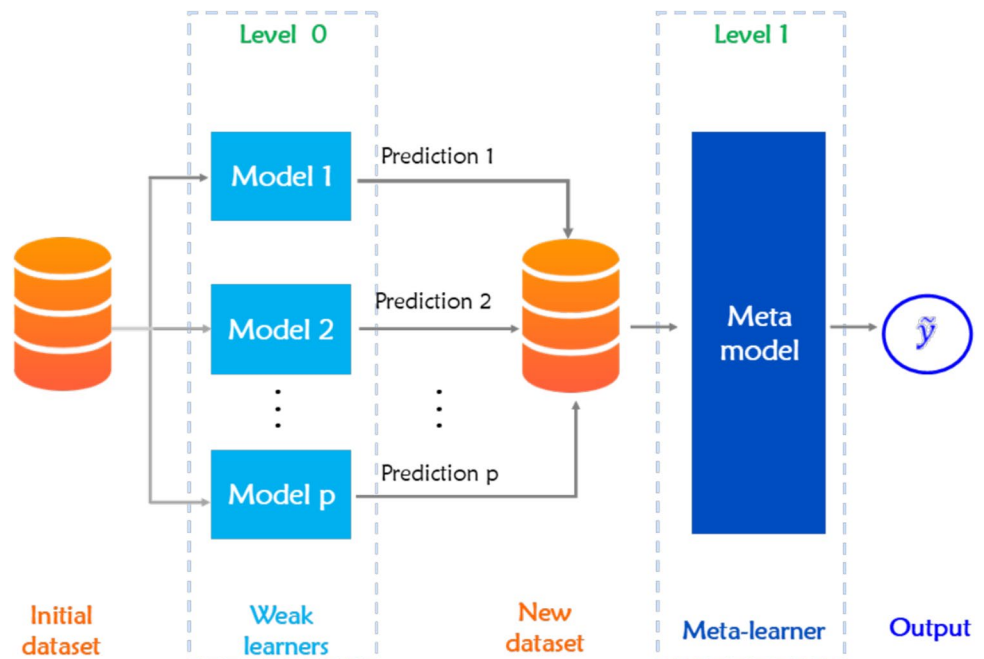
2.1 Background

AI systems have shown very good performance in solving a variety of tasks ranging from the simplest to the most difficult, which has encouraged their adoption in several areas directly affecting people's lives. Ensemble models are one of them, and one type of ensemble model is stacked generalization. The stacked generalization ensemble model (Wolpert, 1992), also called the stacking model (Ting and Witten, 1999), is an ensemble learning model composed of two levels. The first level (level-0) is made up of several machine learning models, which can be of different types, called base learners or base models. The second level (level-1) is made up of a single machine learning model called a meta-learner.

The base learners take as input the same dataset to produce individual predictions, and then the meta-learner combines the predictions produced by these different base learners, taking as input the predictions of the base learners and giving the final prediction. Most often, linear regression is used as the meta-learner. Figure 1 illustrates the architecture of a stacked generalization model.

However, the performance of this type of AI system marred by the opacity of AI models in terms of their decision-making mechanisms, which has earned them the reputation of being black-box models. This opacity undermines the trust that end-users place in AI models. For a model to be adopted by end-users and industries, it must be trustworthy (Véliz et al., 2021). It is therefore necessary to make machine learning models understandable to end users by providing explanations. Providing explanations for a prediction means presenting textual or visual artifacts that provide a qualitative understanding of the relationship between the instance and the model's prediction (Ribeiro et al., 2016). Thus, for an instance of X features, an explanation of the prediction can be expressed as a subset S of features, along with their importance, which contributes mostly to the prediction, ($S \subset X$). Explanations can be categorized as local or global; we often talk about local explainability and global explainability (Bellucci et al., 2021). Local explainability concerns explanations given for one instance of a dataset, while global explainability concerns explanations given for a set of instances, which can be for the whole dataset. To produce explanations, some methods have been developed by researchers, known as explainability methods.

Fig. 1 Stacking architecture



2.2 Literature Review

Explainability Methods Explainability methods can be classified according to different criteria: the complexity of interpretability, the scope of interpretability, and the level of dependency on the used ML model (Adadi and Berrada, 2018). According to the level of dependency on the used machine learning (ML) model, we can distinguish agnostic methods and model-specific methods. An explainability method is qualified as agnostic if it can be applied to any type of model. It is called a model-specific method when the method is designed to be functional on a specific class of AI models.

Among model-specific methods, Layer-wise Relevance Propagation (LRP) (Bach et al., 2015) is a method designed for neural network architectures. Its principle is to redistribute the model's output from the output layer by back-propagation to the input layer. Starting from the output layer, a specific rule is applied layer by layer to obtain the relevance of each neuron until the relevance of the neurons in the input layer is ascertained. The relevance score $R_{i \leftarrow k}^{(l,l+1)}$ of a neuron i of a given layer l on the neurons of layer $l + 1$ is obtained by applying the rule :

$$R_{i \leftarrow k}^{(l,l+1)} = R_k^{(l+1)} \cdot \frac{a_i w_{ik}}{\sum_h a_h w_{hk}} \quad (1)$$

Here, a_i represents the output of neuron i , w_i the weight linking neuron i to neuron k , k is a neuron in layer $l + 1$, and h is the index that covers all neurons in layer l . The final relevance of a neuron i in the lower layer is obtained by summing the relevance from all the neurons k in the upper layer with the rule :

$$R_i^{(l)} = \sum_k R_{i \leftarrow k}^{(l,l+1)} \quad (2)$$

LRP is a stable and robust method due to the conservation property it ensures and the mathematical background on which it is based. However, its limitation is that it can only be used on neural network architectures.

Among agnostic methods, the most popular are SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017) and Local Interpretable Model-agnostic Explanations (LIME) (Ribeiro et al., 2016).

SHAP is a method based on the Shapley value,¹ demonstrated in game theory to be a solution to the problem of fairly sharing the total payoff between several players in a game, according to each player's individual contributions to obtaining this payoff. The SHAP method transposes game

theory into machine learning, where the game is to explain the prediction $f(x)$, the players are the features x_i , and the total payoff is $f(x) - \mathbb{E}[f(x)]^2$; so, the Shapley value corresponds to the contribution of a feature to the prediction $f(x)$. Given a set of features $F = \{1, 2, \dots, p\}$, all variables in a certain coalition $S \subseteq F$ co-operate towards the outcome $val(S)$ with default value $val(\emptyset) = 0$. Shapley values distribute the total value of the $val(F)$ outcome across all features according to their contribution across all possible S coalitions. The marginal contribution of feature i with respect to an S coalition: $\Delta_{val}(i, S) = val(S \cup \{i\}) - val(S)$ is averaged over all $S \subseteq F \setminus \{i\}$. Consequently, the corresponding Shapley values $\phi_{val}(i)$ measure the contribution of i based on Eq. 3.

$$\phi_{val}(i) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(p - |S| - 1)!}{p!} \Delta_{val}(i, S) \quad (3)$$

The high computational time required to calculate Shapley values is a limitation of the SHAP method. To reduce the calculation time, approximate Shapley values are calculated by choosing a subset of feature coalitions. However, considering only some coalitions instead of all often leads to different explanations from one run to another, making the explanations frequently unstable (Kelodjou et al., 2024).

The idea behind the LIME method is to use an interpretable model to locally simulate the complex model. The simple model must already be understandable by humans, like linear regression and decision trees. This simple model, called a surrogate model, uses the neighborhood of the instance to be explained to simulate the behavior of the complex model. The explanation of an instance x produced by LIME is obtained by Eq. 4 :

$$Explanation(x) = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \quad (4)$$

f represents the model to be explained, g the surrogate model used to explain f , π_x is the proximity measure between an instance z and x to define the locality around x , $\Omega(g)$ represents the complexity of g , and $\mathcal{L}(f, g, \pi_x)$ is a measure of the infidelity of g in approximating f in the locality defined by π_x . The aim is to minimize $\mathcal{L}(f, g, \pi_x)$ while ensuring that $\Omega(g)$ is small enough to be interpretable by humans. LIME is an unstable method due to the fact that the neighbors are randomly chosen on each run, which means that for each run, the explanation will be different. Moreover, the quality of the explanation strongly depends on the choice of the neighborhood, whether that choice is good or bad.

¹ Shapley value was introduced by Llyod Shapley (Shapley et al., 1953).

² $\mathbb{E}$ is the mathematical expectation

Explainability of Stacking Model In general, there are several studies within the literature that focus on exploiting stacking models to make predictions. The majority of the studies examine the prediction aspect without addressing the explainability of the predictions obtained by the stacking model (Baradaran Rezaei et al., 2023; Chen et al., 2022, 2017; Elakkiya et al., 2021; Gungor et al., 2022; Hasan et al., 2024; Kang and Kang, 2021; Liao et al., 2022; Liu et al., 2018; Pan and Stark, 2022; Papouskova and Hajek, 2019; Saadatmand et al., 2023; Zhang et al., 2023). Studies on the explainability of ensemble models are relatively recent, and there are many more studies on ensemble models made up of decision trees. However, there are some studies on the explainability of ensemble models of the stacking type (Almohimeed et al., 2023; Hoq et al., 2023; Lu and Qiu, 2023; Nguyen et al., 2021; Zhu et al., 2022).

The work of Almohimeed et al. (2023) proposes a stacking model to predict different classes of Alzheimer's disease and explain its predictions with SHAP and LIME methods. The model is a multi-level stacking composed of three levels: in level 1, six base models (Random Forest (RF), Decision Tree (DT), Support Vector Machine (SVM), Logistic Regression (LR), K-nearest Neighbors (KNN), and Native Bayes (NB)); in level 2, three different linear-regression based models trained on three medical modalities using predictions of base learners in level 1; then at level 3, they merged the predictions of stacking models in level 2 and passed them to the meta-learner (still a linear regression) to produce the final prediction. They utilized SHAP and LIME explainers to understand which features had the most significant impact on model decisions. They used waterfall plot for local explanations and summary plots for global explanations.

Hoq et al. (2023) employ a stacked ensemble model to predict students' final exam grades. This study uses KNN, SVM, and Extreme Gradient Boosting (XGBoost) as the base models and linear regression as a meta-model to combine the predictions of the base models. They used SHAP, to explain the model's predictions to help stakeholders understand the impact of different programming behaviors on student success. They explain the predictions at an individual and a global level. They use force plots for local explanations and summary plots for global explanations.

Lu and Qiu (2023) proposed a stacking ensemble model composed of four base learners (ridge regression, random forest, gradient boosting decision tree, and artificial neural network (ANN)) and a meta-learner (Elastic Net) for predicting the daily number of hospital admissions for cerebrovascular disease. They applied the SHAP method to provide explanations for the predictions on both global and local scales. On the global scale, they used heatmap plots of SHAP values of all features across all samples and also

used SHAP dependence plots. On the local scale, they used waterfall plots of SHAP values.

The stacking model used by Nguyen et al. (2021) consists of a meta-learner (Elastic Net), which combines the predictions of six base learners (Ordinary Least Squares (OLS), Elastic Net, Multi-layer Perceptron, KNN, Random Forest, XGBoost) to predict three levels of CO_2 emissions for corporation. The authors performed explainability on a global scale using the coefficients of linear regression of the base models OLS and Elastic Net to identify the variables that are globally most important for predicting CO_2 emissions. They also obtained feature importance from the random forest and XGBoost models to determine the order and importance value of each feature according to these models. Thus, they used only linear models and decision trees to provide explanations yet the stacking model presented better prediction performance than the individual models.

Zhu et al. (2022) proposed an interpretable stacking-based ML framework to predict concentrations in real time after olanzapine (OLZ) treatment. The stacking ensemble learning framework integrates optimized extra trees, XGBoost, random forest, bagging, and gradient-boosting regressors with linear regression as the meta-learner. Its predictions were explained using LIME and partial dependence plots (PDPs). Explanations at the local level were illustrated by LIME plots, while the global explanations of the relationship between features and output were illustrated by PDPs. The works cited above, except for Nguyen et al. (2021), effectively explain the predictions of the various stacking models designed. We note that agnostic methods are used to explain the predictions of stacking models. Among these agnostic methods, the SHAP method is mostly used in these studies, and like the other methods used, it does not take into account the internal workings of the model to explain the predictions. The approach we are proposing will seek to overcome this limitation. Thus, we aim to propose a method that can provide better explanations by explicitly taking into account the internal architecture of stacking models. This approach is designed to address the limitations of existing agnostic methods by leveraging the hierarchical structure specific to ensemble learning. The proposed method is evaluated using fidelity and stability metrics, and compared to well-known model agnostic explanation techniques such as LIME and SHAP.

3 Method

We start from the LRP method initially designed to explain neural networks and adapt it to explain the predictions of a stacking model. Knowing that in the LRP method, three

elements are used to determine the contribution of a neuron i , namely its output value a_i , its weight w_i and the contributions of the neurons k of the next layer to which it is linked $R_{i \leftarrow k}$ (with k , being a neuron of the next layer). We seek to recover these elements in the stacking model.

We make the following considerations:

- As the base learners are of different types, we are only interested in their outputs.
- The meta-learner produces the final result based on the predictions of the base learners, so there will be weights w_i that link each base learner to the output of the stacking model.
- Since the base learners produce output and have a weight (w_i) that links them to the meta-learner, this is similar to a hidden layer in an ANN or linear regression.

The similarity described above is illustrated in Fig. 2:

We can see that once the similarity has been established, the LRP method can be applied to a stacking model, except that the weights linking the inputs to the base learners will not be known, as there would not be a priori a weight linking each input to each base learner to continue applying the LRP method. We therefore thought of adding a model-agnostic feature importance method to the calculation process to obtain these weights. Furthermore, this explainability method is used between the input layer and the base learner layer. The aim is to calculate the contribution of each variable to the output of each base learner, and this contribution value will be used as the weight w_j which connects each input to each base learner. In this way, the LRP method can be fully applied to a stacking model. Figure 3 gives an overview of the situation described above.

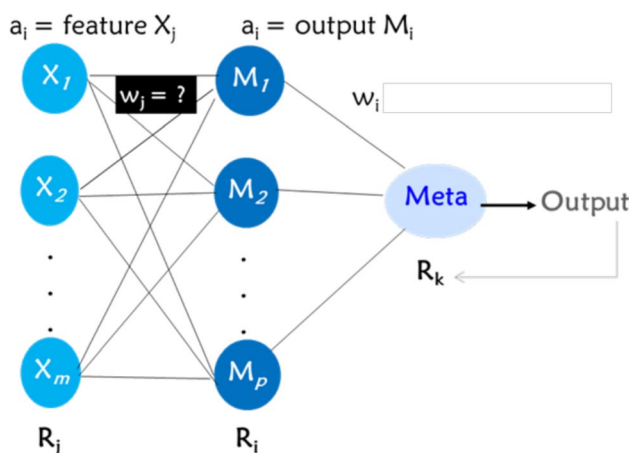


Fig. 2 Similarity ANN vs stacking model

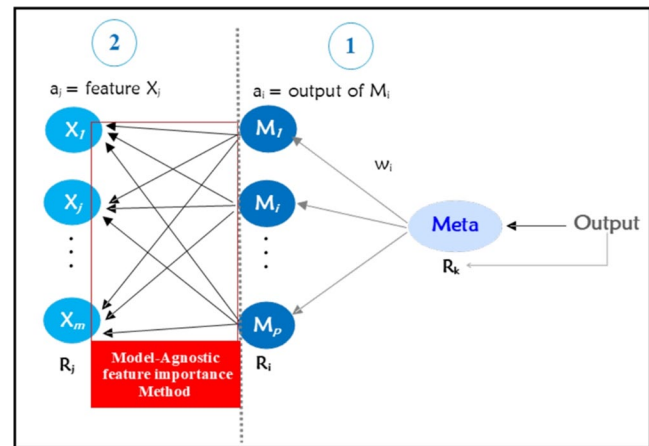


Fig. 3 Proposed method

3.1 General Algorithm

The proposed method, called LSMeta is divided into three steps:

1. Determine the contributions of the base models on the meta-learner output (LRP- Eq. 5),
2. Determine the contributions of the input variables from the contributions of the base models (model-agnostic feature importance method- Eq. 6),
3. Deduct the contributions of each variable to the final output by aggregating the contributions obtained in steps 1 and 2 (LRP- Eq. 7).

Step 1: Determine the contributions of the base models on the meta-learner output The inputs of this step are as follows: the meta-learner output $\tilde{y}$, the coefficients of meta-learner w_i , and the output of each base learner $Pred_i$. The output produced by step 1 is the relevance (contribution) of each base learner to model output R_i . In this step, we will calculate the contribution of each base learner to the $\tilde{y}$ output of the stacking model. To do this, the first propagation rule (Eq. 1) of the LRP method was adapted into Eq. 5. This is translated into line 3 of the algorithm.

$$R_i \leftarrow \tilde{y} * \frac{Pred_i * w_i}{\sum_{h=1}^p Pred_h * w_h} \quad (5)$$

So, LRP rule 1 is applied but using elements proper to a stacking model :

$$\begin{cases} R_i^{(t,l+1)} \equiv R_i : \text{relevance of base learner } i \text{ to model output} \\ R_k^{(t+1)} \equiv \tilde{y} : \text{model output} \\ a_i \equiv Pred_i : \text{output of base learner } i \\ w_{ik} \equiv w_i : \text{coefficient of base learner } i \end{cases}$$

Step 2: Determine the contributions of the variables from the contributions of the base models The inputs of this step are the output of each base learner, feature values, and the contribution of each base learner to the model output (output of step 1). The output produced by step 2 is the relevance (contribution) of each feature to the contribution of each base learner. Here, we will calculate the contribution of each feature to the prediction of each base learner ($feat_imp_{ij}$) using a model-agnostic feature importance method. This method will provides the contributions of the features to the prediction of each base learner. This calculation is done in line 7 of Algorithm 1. The contributions of the features to those of the base learners are calculated by adapting the first LRP propagation rule, but this time into Eq. 6. This is translated into line 10 of Algorithm 1.

$$R_{j \leftarrow i} \leftarrow R_i * \frac{x_j * feat_imp_{ij}}{\sum_{h=1}^m x_h * feat_imp_{ih}} \quad (6)$$

With in this case:

$$\begin{cases} R_{i \leftarrow k}^{(l,l+1)} \equiv R_{j \leftarrow i} : \text{relevance of feature } j \text{ to the contribution of base learner } i \\ R_k^{(l,l+1)} \equiv R_i : \text{relevance of base learner } i \text{ to model output} \\ a_i \equiv x_j : \text{value of feature } j \\ w_{ik} \equiv feat_imp_{ij} : \text{contribution of feature } j \text{ to the output of the base learner } i \end{cases}$$

Step 3: Deduct the contributions of each variable from the final output The inputs of this step are the contributions/relevance of each feature on the prediction of each base learner (output of step 2). The output produced by step 3 is the contribution of each variable to the model output. In this step, each feature's total contribution to the prediction is calculated by adding up its different contributions to each base learner (obtained in step 2). To do this, the second propagation rule (Eq. 2) of the LRP method is adapted into Eq. 7. This is translated into line 15 of Algorithm 1.

$$R_j \leftarrow \sum_{i=1}^p R_{j \leftarrow i} \quad (7)$$

With in this case:

$$\begin{cases} R_i^{(l)} \equiv R_j : \text{relevance of feature } j \text{ to the output} \\ R_{i \leftarrow k}^{(l,l+1)} \equiv R_{j \leftarrow i} : \text{relevance of feature } j \text{ to the contribution of base learner } i \end{cases}$$

These three steps result in Algorithm 1. Step 1 runs from line 1 to 4, then step 2 runs from line 5 to 12. Finally, step 3 runs from line 13 to 16.

Algorithm 1 LSMeta.

Require: black box $M = \{\text{meta, base learners } M_1, \dots, M_p\}$;
 $x(x_1, \dots, x_m) = \text{observation}$; $w_i = \text{coefficients of meta-learner}$;
 $\tilde{y} = \text{prediction to explain}$

```

1: for each base learner  $M_i$  do
2:   Calculate the contribution of base learner  $M_i$  to  $\tilde{y}$  :
3:    $R_i \leftarrow \tilde{y} * \frac{Pred_i * w_i}{\sum_{h=1}^p Pred_h * w_h}$  ▷
   Eq. 5
4: end for
5: for each base learner  $M_i$  do
6:   Calculate the contributions of features of  $x$  to the prediction of  $M_i$  :
7:    $feat\_imp_{ij} \leftarrow \text{featImpMethod}(x, Pred_{M_i})$ 
8:   for each feature  $j$  do
9:     Calculate the contributions of  $j$  to the contribution of  $M_i$ :
10:     $R_{j \leftarrow i} \leftarrow R_i * \frac{x_j * feat\_imp_{ij}}{\sum_{h=1}^m x_h * feat\_imp_{ih}}$ 
    ▷ Eq. 6
11:   end for
12: end for
13: for each feature  $j$  do
14:   Add up the different contributions of  $x_j$  in each base learner  $M_i$ .
   :
15:    $R_j \leftarrow \sum_{i=1}^p R_{j \leftarrow i}$  ▷
   Eq. 7
16: end for
```

Ensure: $R_j = [R_1, \dots, R_m]$

3.2 Properties

Since the proposed method relies on a model-agnostic feature importance technique, its properties are inherently linked to those of the underlying method chosen. In this study, we choose SHAP as the model-agnostic feature importance method, due to its solid theoretical grounding and broad adoption in explainable AI applications (Tchuente et al., 2024). We evaluate the proposed method across six key dimensions: computational complexity, soundness and completeness, categorization among explainability methods, ability to provide global or local explanations, fidelity to the original model, and stability.

Complexity The time complexity C of the proposed algorithm is equal to the sum of the complexities of each step: $C = C_{step1} + C_{step2} + C_{step3}$.

Let m be the number of features in the input example and p the number of base learners making up the stacked generalization model.

Knowing that the sum in the denominator in Eq. 5 will only be done once, $C_{step1} = O(p)$.

$$\begin{aligned}
C_{step1} &= O(p) \\
C_{step2} &= p(C_{feat_imp} + O(m)) \\
C_{step3} &= O(mp) \\
\text{So, } C &= O(p) + p(C_{feat_imp} + O(m)) + O(mp)
\end{aligned}$$

In the case of using SHAP, we have:

$$\begin{aligned}
C_{step2} &= p(C_{shap} + O(m)) \quad \text{yet, } c_{shap} \text{ is exponential (Lundberg \& Lee, 2017)} \\
&= p(O(2^m) + O(m)) \\
&= O(p \cdot 2^m) \\
C &= O(p) + O(p \cdot 2^m) + O(mp) \\
&= O(p \cdot 2^m)
\end{aligned}$$

The proposed method therefore has an exponential complexity, due to the SHAP method in its computational steps. Moreover, this complexity depends on the number of base learners in the stacked generalization model. The computational time of the contributions is proportional to the number of base learners, since the SHAP method must be applied to each of them. If we assume $t_{average}$, the average time taken to calculate contributions for a base learner using the SHAP method for an observation, the time taken to calculate contributions T using the proposed method for an observation will be equal to p times that of the SHAP method: $T = p * t_{average}$

Soundness and Completeness The proposal is based on the well-known algorithms LRP and SHAP. Although the soundness and completeness of these methods are not explicitly discussed in their original papers, these properties can be discussed respectively in terms of their ability to accurately and exhaustively attribute the relevance of inputs. SHAP is based on Shapley values from cooperative game theory, ensuring the model's prediction can be exactly reconstructed from the SHAP values. LRP back-propagates the output in such a way that the total relevance is conserved across layers. This implies that the contribution of input features correctly reflects the model's output. SHAP takes into account all possible combinations of features to determine the contribution of the inputs. LRP relies on all the elements (layers, neurons, activation, weights) involved in the prediction to provide an explanation. These respective arguments ensure that these two methods provide a sound and complete explanation of how each feature affects the prediction. The proposed method, made by the combination of LRP and SHAP, inherits their properties of soundness and completeness. However, in the case of SHAP, as the complete version has exponential time complexity, the authors introduce randomness in the feature selection. This could make it incomplete and unsound.

Agnostic vs Model-Specific We classify the proposed method in the category of model-specific explainability methods (Adadi and Berrada, 2018). The method is

designed to perform on stacked generalization ensemble learning model architecture.

Local vs Global Explainability The method is designed for local explainability but can also be used for global explainability. As done in the SHAP Python library,³ the average of the contributions obtained for local explanations can be used to provide global explanations.

Fidelity In the proposed method, the input contributions are calculated as a function of the contributions of the base learners. These base learner contributions are in turn calculated as a function of the model output. This output relevance decomposition layer-by-layer preserves the structure of the stacking model and ensures that the relevance distribution remains proportionally aligned with the model's decision-making process. Hence, the proposed method would produce explanations more faithful than the associated base learners to the prediction of a stacking model.

Stability As the LRP method is applied to the L1 layer of the stacking model, the relevance scores of this layer will not vary because the same inputs will always produce the same relevance scores. This conservation will reduce the instability of the proposed method even though the SHAP method is subsequently applied to the second layer of the model. Furthermore, once the SHAP method has been applied to the level-1 layer, the LRP method will still be used to calculate the final contributions of the variables.

4 Experiments

The experiment consists of implementing a stacking ensemble learning model, which will serve as a black-box model, then making predictions with the model, followed by explaining the predictions of this black-box model using the proposed method and comparing it with the SHAP and LIME explainability methods. To experiment if taking into account the internal workings of the stacking model effectively improves the explanations, several datasets from different fields were used. Fidelity and stability metrics were used to evaluate the robustness of the proposed method compared to the SHAP and LIME methods.

4.1 Data

The datasets were downloaded from the *UCI* and *Kaggle* platforms, except for the Thompson dataset, which was provided by Thompson Reuters. We used a total of 10 datasets (Table 1),

³ <https://shap.readthedocs.io/en/latest/> accessed on 26 August 2024

Table 1 Datasets summary

	Task	Data type	Instances	Features	NA	Source
Thompson	regression	numeric	14531	52	yes	Thomson Reuters
Abalone	regression	numeric and categorical	4 177	08	no	uci
Concrete	regression	numeric	1 030	08	no	uci
Boston	regression	numeric	545	13	yes	kaggle
Turbine	regression	numeric	36 733	10	no	uci
Bankrupt	classification	numeric	6 819	95	no	uci
Biodeg	classification	numeric	1 050	41	no	uci
Churn	classification	numeric	3 150	13	no	uci
Diabetes	classification	numeric	768	08	no	kaggle
Rice	classification	numeric	3 810	07	no	uci

divided into two groups: the first group deals with regression problems, and the second group deals with classification problems. The datasets dealing with regression problems are as follows :

- *Thompson*: Prediction of CO2 emissions, where a row represents a company and a column contains the value of an environmental, social, or governance characteristic of the company. This dataset contains 14,531 observations and 52 features (target excluded).
- *Abalone*: Prediction of the age of an abalone (marine mollusk) from physical measurements of this mollusk.⁴ This dataset contains 4,177 observations and 08 features.
- *Concrete*: Prediction of the compressive strength of concrete from age and ingredients, in megapascals (MPa).⁵ This dataset contains 1,030 observations and 08 features.
- *Boston*: Prediction of the price of housing in Boston using some housing characteristics, in thousands of dollars.⁶ This dataset contains 545 observations and 13 features.
- *Turbine*: Prediction of the flue gas emissions (CO and NOx) from a gas turbine provide by 11 sensor measures aggregated over one hour, in MWH (megawatt hour).⁷ This dataset contains 36,733 observations and 10 features.

The datasets dealing with classification problems are as follows:

- *Bankrupt*: Determines whether or not the company will go bankrupt, based on the business regulations of the

Taiwan Stock Exchange. This dataset contains 6,819 observations and 95 features.⁸

- *Biodeg*: Classification of chemicals into ready or non-ready biodegradables on the basis of molecular descriptors.⁹ This dataset contains 1,050 observations and 41 features.
- *Churn*: Determines the churn state (churn or non-churn) of the customers at the end of 12 months based on telecom descriptors.¹⁰ This dataset contains 3,150 observations and 13 features.
- *Diabetes*: Determines whether a patient has diabetes or not based on medical measurements.¹¹ This dataset contains 768 observations and 08 features.
- *Rice*: Classifies the species of rice in two classes, Cammeo or Osmancik, based on morphological descriptors obtained for each grain of rice.¹² This dataset contains 3,810 observations and 07 features.

4.2 Experimental Protocol

Preprocessing The following processes were used to prepare the data:

- Deletion of variables with more missing values than the target variable on the *Thompson* dataset;
- Replacement of missing values using the *Multiple Imputation by Chained Equations (MICE)* algorithm (Van Buuren and Groothuis-Oudshoorn, 2011) on the *Thompson* dataset;

⁴ <https://archive.ics.uci.edu/static/public/1/abalone.zip>

⁵ <https://archive.ics.uci.edu/static/public/165/concrete+compressive+strength.zip>

⁶ <https://www.kaggle.com/datasets/yasserh/housing-prices-dataset>

⁷ <https://archive.ics.uci.edu/static/public/551/gas+turbine+co+and+nox+emission+data+set.zip>

⁸ <https://archive.ics.uci.edu/static/public/572/taiwanese+bankruptcy+prediction.zip>

⁹ <https://archive.ics.uci.edu/static/public/254/qsar+biodegradation.zip>

¹⁰ <https://archive.ics.uci.edu/static/public/563/iranian+churn+dataset.zip>

¹¹ <https://www.kaggle.com/datasets/mathchi/diabetes-data-set?resource=download>

¹² <https://archive.ics.uci.edu/static/public/545/rice+cammeo+and+osmancik.zip>

- Z-score normalization of all variables on all the datasets:
 $\forall x, x = \frac{x - \bar{X}}{\sigma}$.

Implementation For the prediction and classification tasks, we used a stacking model. This model was built following the methodology of the study by Nguyen et al. (2021) except for a few details. The stacking model used consists of six base learners: Ordinary Least Squares (OLS), Elastic Net, Random Forest, XGBoost, Multi-layer Perceptron and K-nearest Neighbors. Two strategies were tested to construct the meta-learner: average of the predictions of the base learners and meta-model (OLS and Elastic Net regression); the strategy producing the best performance will be used to carry out explainability. For classification problem, OLS is replaced by logistic regression. We also performed a search for optimal hyper-parameters to obtain the best values for the parameters of the base learners and the meta-model. The following libraries were used for the implementation:

- *Pandas* and *Numpy* for data pre-processing;
- *Matplotlib* for visualizing the results;
- *HyperOpt* and *GridSearch* for the search for optimal hyper-parameters;
- *Scikit-learn* and *XGboost* for implementing the models.

To implement explainability, we used KernelShap (an agnostic variant of the SHAP method) and LimeTabularExplainer (version for tabular datasets of the LIME method) to explain the predictions of all models. To this end, the Python libraries shap and lime were used to obtain explanations and visualizations.

Evaluation The prediction and explanation tasks were evaluated separately. To evaluate prediction, the dataset was divided using the holdout technique, with 80% for the training set and 20% for the test set. For the process of selecting the best hyperparameters, k-order cross-validation (with $k = 10$) was used. The mean absolute error (*MAE*) and the coefficient of determination (R^2) were used as evaluation metrics for the regression problem, while accuracy was used for binary classification. *MAE* evaluates the mean distance between the real and predicted values, R^2 evaluates the quality of the regression, and accuracy measures how the model correctly predicts the outcome.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \tilde{y}_i| \quad (8)$$

$$R^2 = 1 - \frac{\sum (y_i - \tilde{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (9)$$

$$Accuracy = \frac{\text{correct predictions}}{\text{All predictions}} \quad (10)$$

Explainability methods were evaluated with commonly used metrics (Ali et al., 2023; Bellucci et al., 2021; Guidotti, 2021): fidelity and stability. Fidelity to evaluate how accurately an explanation generated by an XAI method reflects the model's predictions, while stability evaluates the capacity of the XAI method to provide identical explanations for the same instance.

Fidelity was evaluated following the approach proposed by Nguyen (2018). This approach consists of determining the most important features V that lead to the prediction $\tilde{y}_i$ for each observation i , then making a new prediction $\tilde{y}_{i \setminus V}$ without these features and calculating the average error between $\tilde{y}_i$ and $\tilde{y}_{i \setminus V}$:

$$fidelity_{score} = \frac{1}{n} \sum_{i=1}^n |\tilde{y}_i - \tilde{y}_{i \setminus V}| \quad (11)$$

The higher this fidelity score is, the more faithful the explanation provided is to the model's prediction. The explanations must be faithful to the model to avoid explaining something other than the results of the model.

To evaluate stability, we used the Variable Stability Index (VSI) proposed by Visani et al. (2022), which evaluates, in a given instance, whether the explanatory variables remain the same after several runs. Let $g_1 \dots g_m$ be the different explanations obtained after each run. We will determine the set of all possible pairs of explanations C_m^2 and determine the number of features common to the models of each pair, which we will divide by the size of the explanation, and finally, we will take the average for all the pairs, which will constitute the stability score. We use two runs to evaluate stability ($m = 2$). The higher the score is, the more stable the explanations are. The stability of explanations is crucial to ensure reliability and confidence in model predictions by providing consistent and reproducible justifications.

In the following subsections, boldface values in the comparative results tables indicate, for each dataset, the best-performing evaluation metric (e.g., model/method with lowest *MAE* or model/method with the highest *Rsquare*) for the dataset considered.

4.3 Regression Problem

Prediction Performance We evaluated the performance of each base learner for the prediction task. These performances are summarized in Tables 2 and 3.

From Tables 2 and 3, we observe that linear regressions (OLS and Elastic Net) produce the worst performance

Table 2 MAE of base learners

	OLS	Elastic Net	KNN	Random Forest	XGBoost	MLP
Thompson	0.356	0.301	0.099	0.123	0.131	0.127
Abalone	0.471	0.471	0.455	0.448	0.490	0.489
Concrete	0.516	0.516	0.349	0.220	0.156	0.155
Boston	0.330	0.330	0.287	0.241	0.258	0.220
Turbine	0.048	0.048	0.040	0.028	0.028	0.024

Table 3 Rsquare of base learners

	OLS	Elastic Net	KNN	Random Forest	XGBoost	MLP
Thompson	0.173	0.129	0.831	0.755	0.818	0.835
Abalone	0.507	0.507	0.515	0.555	0.492	0.509
Concrete	0.608	0.608	0.755	0.906	0.940	0.942
Boston	0.724	0.724	0.677	0.825	0.832	0.833
Turbine	0.996	0.996	0.995	0.998	0.998	0.998

regardless of the dataset. In contrast, the multi-layer perceptron almost always predicts best on most datasets, followed by decision trees and then KNN.

Performances of different combination strategies of base learners were also evaluated to choose the better one for explainability. They are summarized in Tables 4 and 5.

Tables 4 and 5 show that for the majority of datasets, the combination strategy using a simple linear regression (OLS) as the meta-learner obtains better performance in terms of mean absolute error or Rsquare. So, for the rest of the experimentations, we will use a stacking ensemble model with *OLS* as the meta-learner.

Explainability The proposed method LSMeta was implemented and used to explain the predictions made by the stacking model on observations in the test set of various datasets. SHAP and LIME methods were also used to explain predictions of the test set. Each method returns the five most important variables. The contributions made by these methods are then evaluated using measures of fidelity and stability.

Table 4 MAE of combination strategies

	Mean-based	Meta-Elastic Net	Meta-OLS
Thompson	0.161	0.158	0.116
Abalone	0.441	0.668	0.473
Concrete	0.282	0.587	0.155
Boston	0.224	0.478	0.219
Turbine	0.029	0.391	0.023

Table 5 Rsquare of combination strategies

	Mean-based	Meta-Elastic Net	Meta-OLS
Thompson	0.758	0.739	0.856
Abalone	0.571	0.111	0.549
Concrete	0.873	0.552	0.942
Boston	0.850	0.488	0.872
Turbine	0.998	0.711	0.998

Some Local Explanations The proposed method provides local explanations, meaning that for a given instance, it returns the contribution of each variable x_i to the prediction $\hat{y}$ of the model. For instance, Figs. 4 and 5 below present the data for two randomly selected instances from the Concrete and Boston datasets, respectively.

Figures 6 and 7 respectively present a local explanation provided by the proposed method of each instance of Figs. 4 and 5 above.

In Fig. 6, we observe that the most important feature is *Cement*, followed by *Superplasticizer*, indicating that those features strongly contribute to increasing the predicted compressive strength, while *water* contribute to reducing it. Therefore, to obtain a high compressive strength of concrete, it is necessary to adjust or increase the values of *Cement* and *Superplasticizer*. Conversely, if a lower compressive strength is desired, one should increase the values of *water*.

In Fig. 7, we observe that the most important feature is *RM* (*Average number of rooms per dwelling*), followed by *RAD* (*Index of accessibility to radial highways*), indicating that *RM* and *RAD* strongly contribute to increasing the predicted house price, while *PTRATIO* (*Pupil-teacher ratio by town*) and *LSTAT* (*% lower status of the population*) contribute to decreasing it. Therefore, to achieve a high house price, it is necessary to adjust or increase the value of *RM* and *RAD*. Conversely, if a lower price is expected, the value of *PTRATIO* and *LSTAT* must be increased.

General Evaluation First, for an overview of performance, we calculate the average fidelity scores by averaging the individual fidelity scores of each instance. The results for SHAP, LIME and the proposed method are shown in Table 6.

It can be observed that, for each dataset, the average fidelity score of all three methods is significantly different from zero, indicating that they produce faithful explanations. However, LIME consistently yields lower fidelity scores compared to the other two methods. Concerning SHAP

Cement	Blast_Furnace_Slag	Fly_Ash	Water	Superplasticizer	Coarse_Aggregate	Fine_Aggregate	Age	Concrete_compressive_strength
1.376969	0.375755	-0.847144	-1.315006	1.724241	-1.554617	1.416567	-0.612331	0.801433

Fig. 4 An example of instance in the Concrete dataset with $\tilde{y} = 0.801$. The last column corresponds to the target value

CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
-0.378616	0.356941	-1.111785	-0.271237	-0.976512	0.763141	-1.318794	0.140957	-0.510706	-1.132855	-1.681171	0.43074	-1.084936	1.395229

Fig. 5 An example of instance in the Boston dataset with $\tilde{y} = 1.395$

Fig. 6 A local explanation in Concrete dataset

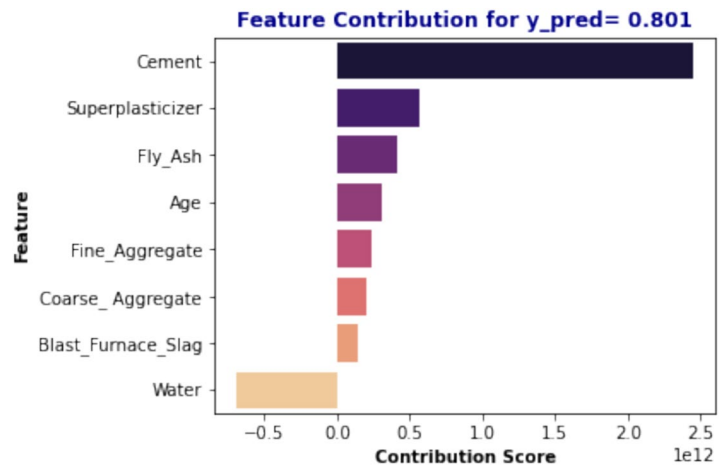


Fig. 7 A local explanation in Boston dataset

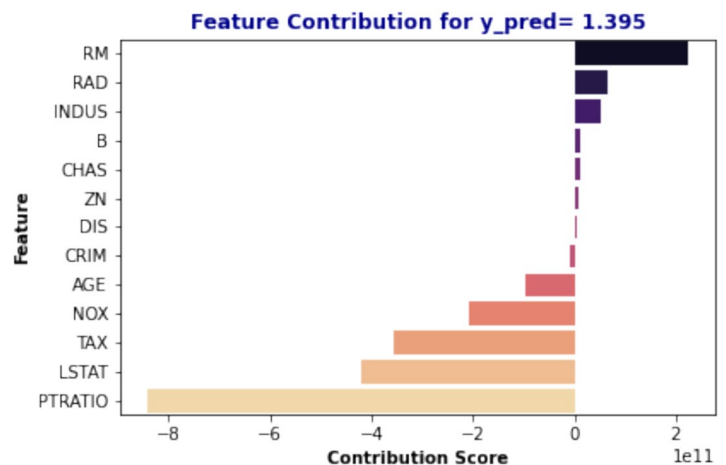


Table 6 Average fidelity scores

	SHAP	LIME	LSMeta
Thompson	0.216 (± 0.74)	0.202 (± 0.69)	0.096 (± 0.25)
Abalone	0.730 (± 0.70)	0.663 (± 0.46)	1.909 (± 1.51)
Concrete	0.559 (± 0.48)	0.571 (± 0.49)	0.624 (± 0.45)
Boston	0.597 (± 0.69)	0.547 (± 0.64)	0.350 (± 0.40)
Turbine	0.457 (± 0.60)	0.414 (± 0.57)	0.447 (± 0.55)

and LSMeta, the average fidelity of LSMeta is greater on the *Abalone* and *Concrete* datasets, while for SHAP it is greater on the *Thompson*, *Boston*, and *Turbine* datasets. The

mean value can be strongly influenced by outliers, and the standard deviation denotes that individual fidelity scores of instances are not very close to the average fidelity score, since all standard deviation values are greater than 0.25. Thus, we think that it would be incomplete to compare the two methods only by using the mean of fidelity scores.

Hence, we explore the fidelity score for each prediction. We determine the number of instances in which each method provides a more faithful explanation than the other. The proportion of instances in the test set where each method has an individual fidelity score higher than the other for various

datasets is summarized in Table 7. In the column related to LSMeta, the values on the left correspond to the comparison between LSMeta and SHAP, while those on the right refer to the comparison between LSMeta and LIME.

Compared to LIME, LSMeta exhibits a higher proportion of fidelity scores across most datasets (Abalone, Concrete and Turbine). Although SHAP achieves a higher proportion of fidelity scores on the majority of datasets, LSMeta outperforms it on two datasets (Abalone and Concrete). In the case of the Abalone dataset for instance, LSMeta provides explanations with higher fidelity scores for 79% of the instances, compared to only 21% for SHAP. For the same dataset, LSMeta also outperforms LIME (67% compared to 33%). This demonstrates that, in the context of local explanations, the proposed method can, in certain situations, provide more faithful explanations than SHAP and LIME.

Since the beginning of the experiments, we set the number of important features to be returned by both methods at five. In the following, we test different explanation sizes (number of important features to be returned) and observe the evolution of individual fidelity scores and stability scores. Empirically, we successively set the size of the explanation to one feature, then $\frac{1}{10}$ to the total number of features, then $\frac{1}{4}$, $\frac{1}{3}$, and finally $\frac{1}{2}$. Tables 8 and 9¹³ show the percentage of instances where LSMeta achieves a higher individual fidelity score than SHAP and LIME, respectively, across different explanation sizes. Figure 8 illustrates Tables 8 and 9 in terms of fidelity.

In terms of fidelity, the results vary across datasets when compared to LIME and SHAP, although a similar overall trend is observed in both tables.

- For the *Thompson* dataset, as the number of important features to be returned increases, the fidelity of the proposed method is consistently greater than or equal to that of SHAP and, respectively, LIME.
- For the *Abalone* dataset, the fidelity percentage of the proposed method initially increases and then decreases when compared to LIME, while it decreases and then increases when compared to SHAP. Nevertheless, it consistently remains higher than that of both SHAP and LIME across the dataset.
- Regarding the *Concrete* dataset, the percentage of fidelity of the proposed method initially decreases, then increases as the number of important features increases.

¹³ In row, the percentage of instances in which LSMeta has a fidelity/stability score greater than or equal to that of SHAP, per dataset. In columns, the percentage obtained for each explanation size following the dataset.

Table 7 Proportion of individual fidelity scores with the five most important variables

Higher Fidelity percentage	SHAP	LSMeta	LIME
Thompson	59%	41% 39%	61%
Abalone	21%	79% 67%	33%
Concrete	44%	56% 59%	41%
Boston	77%	23% 27%	73%
Turbine	66%	34% 59%	41%

Table 8 Comparison of the percentage of instances in which the individual fidelity score obtained for the LSMeta method exceeds that of SHAP for various explanation sizes - Regression task

LSMeta>=SHAP (%)	1	1/10	1/4	1/3	1/2
Thompson	34.25	41.07	55.62	58.17	65.67
Abalone	80.50	80.50	69.38	73.09	77.87
Concrete	66.02	66.02	44.17	49.51	56.31
Boston	46.84	26.58	25.32	22.78	22.78
Turbine	85.11	85.11	60.69	48.80	39.98

Table 9 Comparison of the percentage of instances in which the individual fidelity score obtained for the LSMeta method exceeds that of LIME for various explanation sizes - Regression task

LSMeta>=LIME (%)	1	1/10	1/4	1/3	1/2
Thompson	36.15	39.04	46.44	49.85	60.44
Abalone	61.60	61.60	67.94	69.38	66.75
Concrete	63.59	63.59	31.07	41.75	48.06
Boston	50.63	29.11	26.58	26.58	25.32
Turbine	82.37	82.37	77.99	60.54	58.58

- Finally, for the *Boston* and *Turbine* datasets, an increase in the number of important features results in a decrease in the fidelity percentage, relative to SHAP and LIME, respectively.

Tables 10 and 11 present the percentage of instances in which the individual stability score obtained by the LSMeta method exceeds that of SHAP and LIME, across different explanation sizes.

In terms of stability, the trend is consistent across all datasets: regardless of the number of important variables to be identified. The proposed method exhibits stability that is equal to or higher than that of SHAP in 100% of the observations, except for *Thompson*, where it remains above 96%. The same pattern is observed when comparing the proposed method with LIME (Fig. 9).

4.4 Classification Problem

Prediction Performance Table 12 shows the accuracy of each base learner on each dataset for the classification task.

We can observe that there are no base learners that maintain better performance across all the datasets. For each

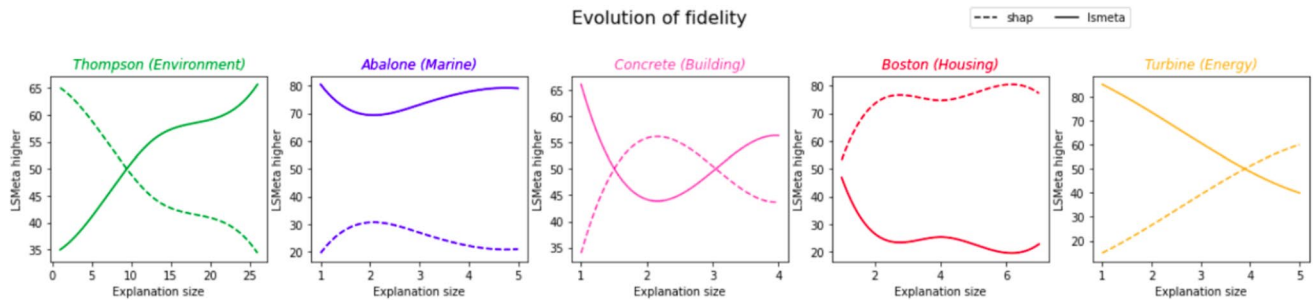


Fig. 8 Comparison of the evolution of the percentage of individual fidelity score (LSMeta vs SHAP) - Regression

Table 10 Comparison of the percentage of instances in which the individual stability score obtained for the LSMeta method exceeds that of SHAP for various explanation sizes - Regression task

LSMeta >= SHAP (%)	1	1/10	1/4	1/3	1/2
Thompson	97.83	98.42	98.38	98.31	97.76
Abalone	100	100	100	100	100
Concrete	100	100	100	100	100
Boston	100	100	100	100	100
Turbine	100	100	100	100	100

Table 11 Comparison of the percentage of instances in which the individual stability score obtained for the LSMeta method exceeds that of LIME for various explanation sizes - Regression task

LSMeta >= LIME (%)	1	1/10	1/4	1/3	1/2
Thompson	96.80	97.25	98.52	98.56	98.45
Abalone	100	100	100	100	100
Concrete	100	100	100	100	100
Boston	100	100	100	100	100
Turbine	100	100	100	100	100

dataset, a different model outperforms the others, except for *KNN* and *XGBoost*, which are the best for two datasets.

The performance of combination strategies is summarized in Table 13.

We observe that the combination strategy using a simple logistic regression as the meta-learner obtains better accuracy for most of the datasets. So, for explainability, we use a stacking model with logistic regression (*log.Reg.*) as the meta-learner.

Explainability: Some Local Explanations Figures 10 and 11 below present the data for two randomly selected instances from the Diabetes and Churn datasets, respectively.

Figures 12 and 13 respectively present the local explanations generated by the proposed method for the instances shown in Figs. 10 and 11.

In Fig. 12, we observe that the most important feature is *Pregnancies*, followed by *Glucose*, indicating that these features strongly contribute to increasing the probability of diabetes predicted by the model. In contrast, *BloodPressure* contributes to reducing the predicted probability. So, pregnancies and glucose levels are associated with an increased risk of diabetes, whereas blood pressure values are linked to a decreased predicted probability of diabetes in the model.

In Fig. 13, we observe that the most important feature is *Complains*, followed by *Call Failure*, indicating that both strongly contribute to an increased probability of customer churn. This suggests that customers who frequently report issues or experience call failures are more likely to discontinue the service. Conversely, features such as *Frequency of Use* and *Subscription Length* have a negative contribution to churn, meaning that customers who use the service more frequently or have longer subscriptions are less likely to leave.

General Evaluation We directly tested different explanation sizes (number of important features) and observed the evolution of individual fidelity and stability scores as we did for regression problems. Tables 14 and 16 show the percentage of instances in which the individual fidelity and stability

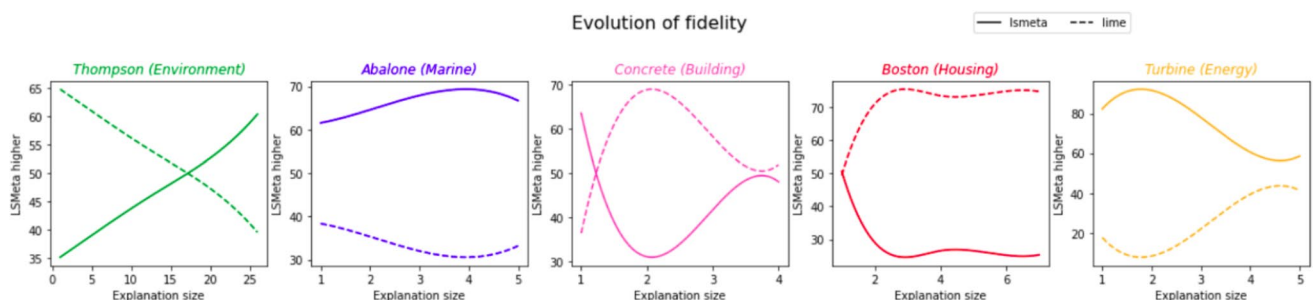


Fig. 9 Comparison of the evolution of the percentage of individual fidelity score (LSMeta vs LIME) - Regression

Table 12 Accuracy of base learners

	Log Reg.	Elastic Net	KNN	Random Forest	XGBoost	MLP
Bankrupt	0.963	0.967	0.965	0.968	0.969	0.969
Biodeg	0.857	0.862	0.905	0.881	0.872	0.872
Churn	0.900	0.903	0.966	0.965	0.973	0.950
Diabetes	0.798	0.798	0.746	0.779	0.772	0.772
Rice	0.922	0.922	0.929	0.923	0.921	0.919

Table 13 Accuracy of combination strategies

	Mean-based	Meta-Elastic Net	Meta-Log.Reg.
Bankrupt	0.968	0.965	0.967
Biodeg	0.900	0.872	0.886
Churn	0.958	0.963	0.965
Diabetes	0.798	0.785	0.779
Rice	0.925	0.923	0.923

score obtained for the LSMeta method are greater than those of SHAP for each explanation size. Similarly, Tables 15 and 17 report the same comparison between LSMeta and LIME. Figures 14, 15, 16 and 17 illustrate those tables, in terms of fidelity and stability respectively.

Concerning the evolution of the percentage of observations where LSMeta has a better fidelity score according to the size of the explanation to be returned, the evolution is not the same for all the datasets. Concerning comparison with SHAP, the percentage tends to decrease for the *Biodeg* and *Rice* datasets, while for the *Churn* dataset, there is an increasing trend. For the *Bankrupt* and *Diabetes* datasets, there is a non-monotonic trend. For the *Bankrupt* dataset, we observe a slight decrease in this percentage (of the order of 0.1) then it rises to 100% for 1 out of 3 and 1 out of 2 important values to be returned. For the *Diabetes* dataset, we observe a growth of this percentage up to 100% then it goes back down to 99.35% for 1 out of 2 important values to be returned.

Concerning the comparison with LIME, as the size of the explanations increases, the percentage decreases for the *Rice* dataset, it decrease then increase for *Biodeg*, and the opposite trend (an increase followed by a decrease) for *Diabetes*. For the *Bankrupt* and *Churn* datasets, we observe the same trend as in the comparison with SHAP.

With regard to stability, unlike in regression problems, we observe a general downward trend: as the size of the explanation increases, the percentage of observations where the proposed method outperforms SHAP and LIME decreases. An exception is the *Bankrupt* dataset, where this percentage remains consistently high—fluctuating between 99.93% and 100% when compared to SHAP, and between 99.78% and 99.73% when compared to LIME. Nevertheless, this decline is not very large, as the percentage remains above 90% for all datasets except for LIME in one case, where it drops to 68%.

Overall, in terms of both fidelity and stability, it can be observed that LSMeta produces more stable and faithful explanations than SHAP for at least 90% of observations for all the datasets.

The results for classification and regression problems highlight the fact that, compared to an agnostic method such as SHAP or LIME, applying the proposed explainability method that takes into account the inner workings of the stacking model can improve explanations, certainly for the stability property and potentially for the fidelity property. We believe that this

Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
2.718712	0.973512	1.0802	0.781814	-0.441087	-0.659038	0.782584	0.830381	1

Fig. 10 An example of instance in the Diabetes dataset with $\tilde{y} = 1$. The last column corresponds to the target value

Call Failure	Complains	Subscription Length	Charge Amount	Seconds of Use	Frequency of use	Frequency of SMS	Distinct Called Numbers	Age Group	Tariff Plan	Status	Age	Customer Value	Churn
-1.050285	3.474268	-2.513019	0.037573	-0.764659	-0.948722	-0.625335	-0.84288	0.194941	-0.290409	-0.574662	-0.113074	-0.788985	1

Fig. 11 An example of instance in the Churn dataset with $\tilde{y} = 1$

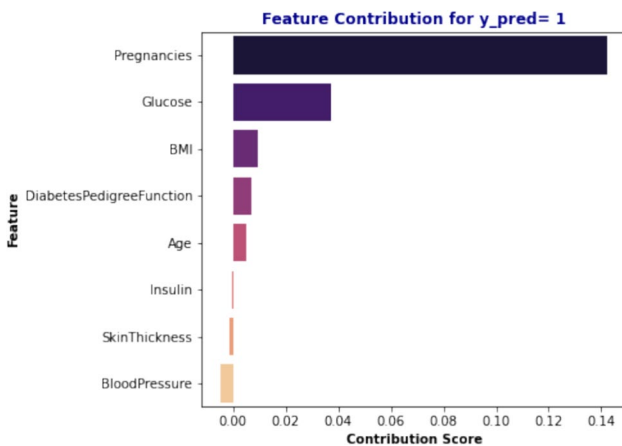
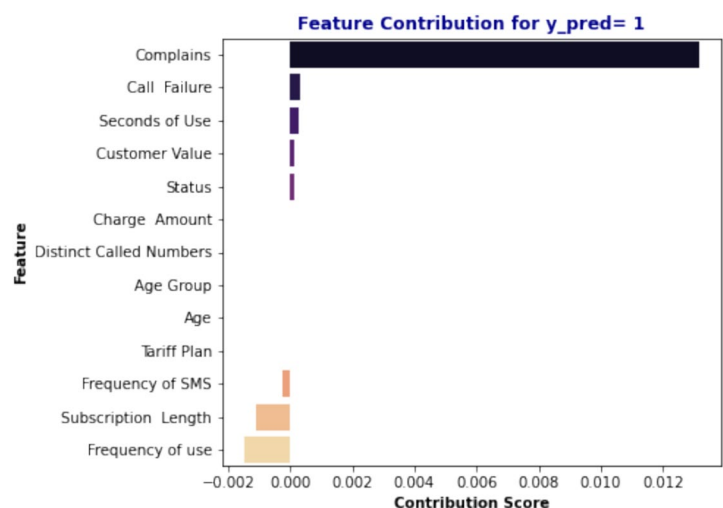


Fig. 12 A local explanation in Diabetes dataset

improvement is likely due to the random choice of feature combinations. Indeed, the LRP method used in the proposed method guarantees that the first level of contribution will be preserved regardless of the execution because the contributions of the base learners do not change. Thus, although the SHAP method is used subsequently, the instability effect will be partially mitigated by the conservation of contributions provided by LRP, which would lead to more stable explanations.

Fig. 13 A local explanation in Churn dataset



5 Discussion

5.1 Theoretical and Practical Implications

Theoretical Implications The explainability method proposed in this study aims to explain the predictions of a stacking model by taking into account the internal operation of this type of model. From a theoretical point of view, the method differs from other explainability methods in that it combines two categories of methods: a model-specific and an agnostic method, LRP and SHAP (or LIME), respectively. Using LRP allows the method to follow the operation of the stacking model, in the sense that the contributions of layer l of the stacking model will be calculated based on the contributions of the following layer $l + 1$ while using an agnostic method such as SHAP will make it possible to manage the diversity of base models in the sense that it allows explaining the individual prediction of each base model independently.

One of the advantages of such an approach is that the internal workings of the stacking model are taken into account in the calculation of contributions, which ensures that the explanation provided is not only based on the model's prediction but also draws on the process that led to

Table 14 Comparison of the percentage of instances in which the individual fidelity score obtained for the LSMeta method exceeds that of SHAP for various explanation sizes - Classification task

LSMeta>=SHAP (%)	1	1/10	1/4	1/3	1/2
Bankrupt	100	99.78	99.93	100	100
Biodeg	98.10	94.79	91.47	90.52	90.05
Churn	94.92	96.03	98.10	98.41	99.37
Diabetes	97.40	97.40	99.35	100	99.35
Rice	99.21	99.21	96.72	95.28	93.83

Table 15 Comparison of the percentage of instances in which the individual fidelity score obtained for the LSMeta method exceeds that of LIME for various explanation sizes - Classification task

LSMeta>=LIME (%)	1	1/10	1/4	1/3	1/2
Bankrupt	100	99.78	99.93	100	100
Biodeg	98.58	96.68	89.10	90.05	90.52
Churn	96.35	97.14	99.37	99.21	99.52
Diabetes	98.70	98.70	98.70	99.35	98.70
Rice	99.61	99.61	97.70	97.24	95.41

this prediction. The result is an explanation that is closely aligned with the model's internal workings. Another advantage is that the proposed method manages the heterogeneity of the base learners in that whatever the model class of the base learners, the method can be used.

Practical Implications The stacking model is an ensemble learning model that improves prediction performance compared with an individual model by combining predictions from different models. This combination of heterogeneous and varied models gives the stacking model the advantage of being able to capture different aspects of the data, having a better capacity for generalizing predictions, and being flexible in that any type of model can be used. All these elements contribute to improving the prediction performance of the stacking model, which is why it is widely used in the literature to solve complex problems in several fields (Almohimeed et al., 2023; Baradaran Rezaei et al., 2023; Chen et al., 2022, 2017; Elakkiya et al., 2021; Gungor et al., 2022; Hasan et al., 2024; Hoq et al., 2023; Kang and Kang, 2021; Liao et al., 2022; Liu et al., 2018; Lu and Qiu, 2023; Nguyen et al., 2021; Pan and Stark, 2022; Papouskova and Hajek, 2019; Saadatmand et al., 2023; Zhang et al., 2023; Zhu et al., 2022). Although widely used for its predictive performance, the explainability aspect of the stacking model

Table 17 Comparison of the percentage of instances in which the individual stability score obtained for the LSMeta method exceeds that of LIME for various explanation sizes - Classification task

LSMeta >=LIME (%)	1	1/10	1/4	1/3	1/2
Bankrupt	99.93	99.78	99.78	99.78	99.93
Biodeg	92.89	83.41	69.67	68.25	68.72
Churn	98.41	97.62	96.35	95.08	92.22
Diabetes	99.35	99.35	97.40	96.75	93.51
Rice	99.61	99.61	97.77	94.62	94.62

is still little explored. This study proposes a method that can be used for the explainability of the stacking model with the particularity of taking into account its specific structure.

In practice, once the stacking model has been built and evaluated, the proposed method can be used to explain the predictions of the stacking model. The practical benefits of using the proposed method can be seen in different profiles: (i) The data scientist (technical expert): gaining a better understanding of stacking models, which can help optimize their results throughout the processing chain (from data collection to mastery of the models created). (ii) The business expert: having a clearer understanding of model predictions fosters adoption and enables the faster identification and correction of biases. (iii) Business managers: gaining greater confidence in the use of complex, high-performance models such as stacking in their businesses.

5.2 Limitations and Futures Research Directions

The proposed method is designed to be applicable to any stacking ensemble model, regardless of the choice of meta-learner. However, the experiments conducted in this study focus exclusively on the case where the meta-learner is a linear model. When linear regression is used as the meta-learner, the weights w_i associated with each base learner naturally correspond to the regression coefficients, thereby providing a direct and interpretable measure of each base model's contribution. In contrast, for nonlinear meta-learners, these weights w_i are not explicitly defined and must be estimated through alternative means. To address this challenge, several avenues could be explored in future work. First, the nonlinear meta-learner could be approximated by an equivalent linear model to enable the extraction of interpretable coefficients, see for example, Rahimi and Recht (2007) for the use of random feature mappings to linearize nonlinear models. Second, the

Table 16 Comparison of the percentage of instances in which the individual stability score obtained for the LSMeta method exceeds that of SHAP for various explanation sizes - Classification task

LSMeta>=SHAP (%)	1	1/10	1/4	1/3	1/2
Bankrupt	100	99.93	99.93	100	99.93
Biodeg	97.63	96.68	94.79	94.31	91.00
Churn	98.10	97.94	96.98	96.35	95.87
Diabetes	99.35	99.35	97.40	96.10	94.81
Rice	99.61	99.61	97.51	93.96	91.99

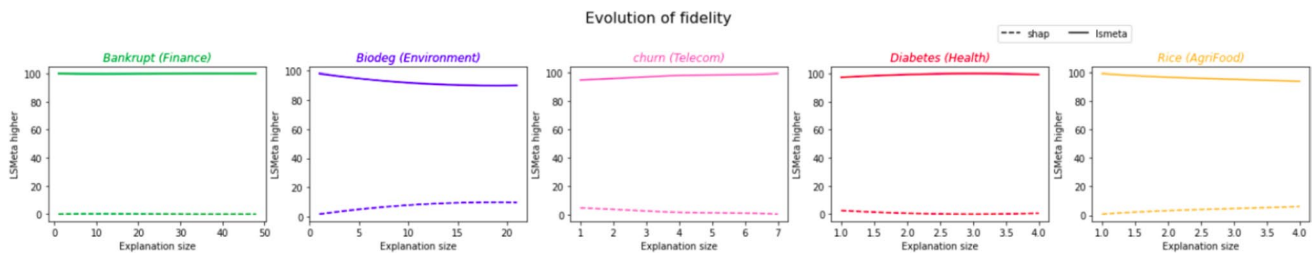


Fig. 14 Comparison of the evolution of the percentage of individual fidelity score (LSMeta vs SHAP) - Classification

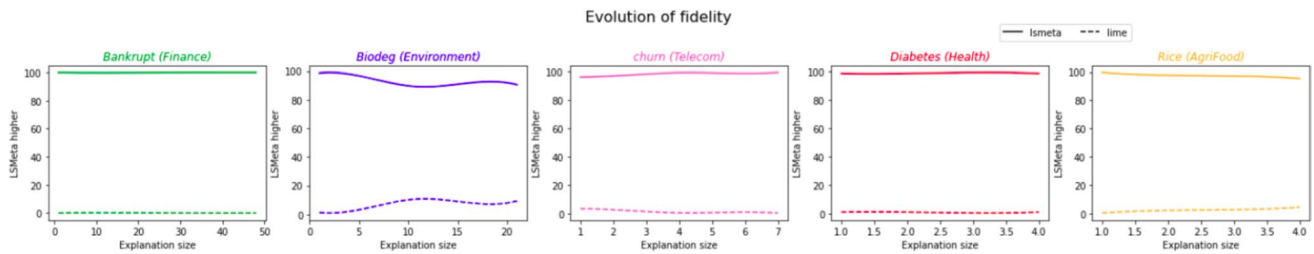


Fig. 15 Comparison of the evolution of the percentage of individual fidelity score (LSMeta vs LIME) - Classification

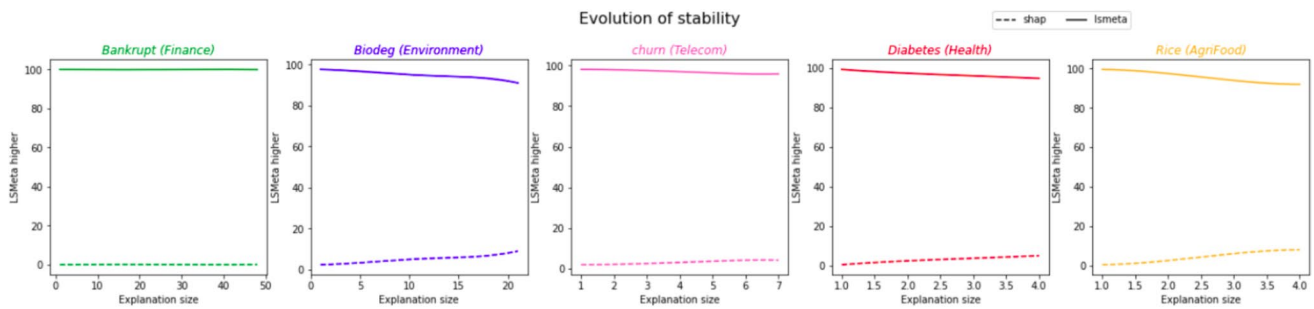


Fig. 16 Comparison of the evolution of the percentage of individual stability score (LSMeta vs SHAP) - Classification

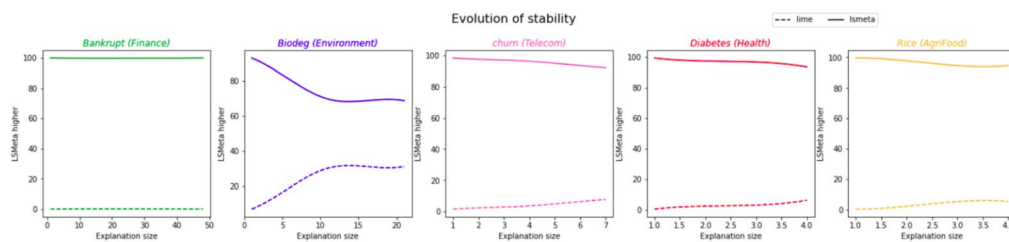


Fig. 17 Comparison of the evolution of the percentage of individual stability score (LSMeta vs LIME) - Classification

internal mechanisms of the nonlinear meta-learner could be analyzed using post-hoc explainability methods to estimate the relative contribution of each base learner to the final prediction. Common techniques such as SHAP, LIME, or other model agnostic methods offer a principled way to approximate these importance scores which can be used as coefficients for non-linear meta-learners in the proposed method.

The computational complexity of the proposed method is primarily determined by the model-agnostic feature importance technique employed in Step 2 of the algorithm. In this study, we mainly used SHAP to evaluate feature importance, which is known to have exponential complexity in the general case. However, this limitation can be mitigated in practice by leveraging SHAP approximations that are tailored

to specific types of base learners. For instance, TreeSHAP, an efficient implementation for tree-based models, offers a substantial reduction in computational cost compared to the generic KernelSHAP approach. Furthermore, if an alternative method such as LIME is used, which operates with linear complexity, the overall computational burden of the proposed method would similarly be reduced to linear time. Thus, the choice of explanation technique plays a critical role in determining the scalability of the method.

Finally, this study presents a rigorous quantitative evaluation conducted across multiple datasets from diverse domains, supplemented by exploratory analyses of instance-level explanations. However, as highlighted by Tchuente et al. (2024), an important direction in the field of explainable AI is to complement quantitative evaluations with human-centered assessments. These involve qualitative evaluations that actively engage end-users, incorporating their feedback, perceptions, and domain-specific expertise to assess the relevance, usefulness, and trustworthiness of the explanations. Integrating such user-centric perspectives constitutes a critical avenue for future research in the general context of the proposed method of this study where interpretability and user trust are essential for practical adoption. Another interesting user-related research direction would be to enrich the proposed method by integrating counterfactual explanations. This approach would make it possible to generate alternative scenarios to better assess the impact of modifications on the model's predictions. Offering a more detailed view of influential factors would provide users with more precise and directly exploitable insights, making it easier to adjust strategies and understand potential biases present in the model.

6 Conclusion

The stacking model is an ensemble learning approach whose inherent complexity necessitates an explanation of its decision-making mechanism. Our work consisted of explaining the results provided by a stacking model by taking into account its internal operation. To address this explainability challenge on stacking models, we proposed a dedicated hybrid explanation method that combines LRP with a model-agnostic feature importance technique. SHAP was chosen as the model-agnostic component for implementing the proposed approach. We evaluated the method against SHAP and LIME on both regression and classification tasks. In regression problems, the proposed method consistently produced more stable explanations across 100% of the observations, irrespective of the explanation size. In terms of fidelity, the performance of the proposed method varied with the explanation size, outperforming SHAP and LIME in some cases,

and underperforming in others. For classification tasks, the proposed method achieved higher stability and fidelity than SHAP in at least 90% of the observations, depending on the explanation size. Compared to LIME, the proposed method generated more faithful explanations for at least 89% of the observations and more stable explanations for at least 68%.

Beyond these results, the work conducted provides numerous and significant directions for future research, particularly in addressing stacking models with nonlinear meta-learners, reducing the computational complexity of the proposed method, and conducting user-centered evaluations as the ultimate means of validating the approach. Lastly, the environmental impacts of such highly complex models must also be carefully considered (Delanoë et al., 2023).

Data Availability The datasets used in this paper are publicly available from links provided in Section 4.1

Declarations

Conflicts of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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